Lesson 2: Understanding Nutrients – Macronutrients and Micronutrients

Holistic Human Nutrition Certificate Program

Lesson Overview

Now that you understand the philosophy of holistic nutrition, it is time to learn about the nutrients that make life possible.

Every heartbeat, every breath, every thought, every movement, and every healing process depends upon nutrients. The human body requires dozens of nutrients every day to maintain health, produce energy, repair tissues, regulate hormones, and support the immune system.

These nutrients are divided into two primary groups:

- **Macronutrients**
- **Micronutrients**

Both are essential for life, but they serve different purposes.

This lesson introduces the major nutrient classes and explains how each contributes to whole-body wellness.

Learning Objectives

Upon completion of this lesson, students should be able to:

- Define macronutrients.
- Define micronutrients.
- Identify the three macronutrients.
- Explain the functions of carbohydrates, proteins, and fats.
- Describe the importance of vitamins and minerals.
- Understand nutrient balance.
- Recognize how deficiencies may affect wellness.

What Are Nutrients?

Nutrients are substances found in food that the body needs for:

- Growth
- Energy
- Repair
- Cellular function
- Hormone production
- Immune support
- Brain function
- Reproduction
- Healing
- Overall wellness

Without nutrients, the body cannot properly perform its daily functions.

Every cell relies upon an adequate supply of nutrients to survive.

Two Major Categories of Nutrients

Human nutrition is generally divided into:

Macronutrients

These nutrients are required in **large amounts** because they provide energy and structural support.

They include:

- Carbohydrates
- Proteins
- Fats

Micronutrients

These nutrients are required in **smaller amounts**, yet they are absolutely essential.

Micronutrients include:

- Vitamins
- Minerals

Although they do not provide calories, they help regulate nearly every biochemical process in the body.

Macronutrients

Macronutrients provide the fuel that powers the body.

Each one performs unique and important functions.

Carbohydrates

Primary Function

Carbohydrates are the body's preferred source of energy.

During digestion, carbohydrates are broken down into glucose, which is used by cells for energy.

The brain, nervous system, muscles, and red blood cells rely heavily on glucose to function properly.

Types of Carbohydrates

Simple Carbohydrates

These digest quickly.

Examples include:

- Fruit sugars
- Milk sugars
- Table sugar

Complex Carbohydrates

These digest more slowly and often contain fiber.

Examples include:

- Brown rice
- Oats
- Sweet potatoes

- Beans
- Lentils
- Whole grains

Complex carbohydrates provide longer-lasting energy and help support digestive health.

Dietary Fiber

Fiber is a special type of carbohydrate that is not completely digested.

Fiber helps support:

- Healthy digestion
- Regular bowel movements
- Blood sugar balance
- Satiety
- Healthy cholesterol levels

Whole plant foods are excellent sources of fiber.

Protein

Primary Function

Protein is the body's primary building material.

Proteins are necessary for:

- Muscle development
- Tissue repair
- Hormone production
- Enzyme production
- Immune function
- Skin
- Hair
- Nails
- Organ function

Every cell in the body contains protein.

Amino Acids

Proteins are made from smaller building blocks called **amino acids**.

Some amino acids are produced by the body.

Others must come from food and are called **essential amino acids**.

A well-balanced diet helps provide all of the amino acids necessary for healthy body function.

Protein Sources

Examples include:

Animal Sources

- Eggs
- Poultry
- Fish
- Lean meats
- Dairy products

Plant Sources

- Beans
- Lentils
- Chickpeas
- Peas
- Soy foods
- Nuts
- Seeds

A variety of protein-rich foods helps support tissue repair and overall wellness.

Healthy Fats

Fat is often misunderstood.

Healthy fats are essential for life.

The body depends upon fats for many important functions.

Functions of Healthy Fats

Healthy fats support:

- Energy storage
- Cell membranes
- Hormone production
- Brain function
- Nervous system health
- Vitamin absorption
- Organ protection
- Body temperature regulation

Certain vitamins cannot be absorbed without dietary fat.

Healthy Fat Sources

Examples include:

- Olive oil
- Avocados
- Walnuts
- Almonds
- Flaxseed
- Chia seed
- Pumpkin seeds

Moderation and quality are key principles in choosing dietary fats.

Micronutrients

Micronutrients are required in much smaller amounts than macronutrients, but they are involved in thousands of chemical reactions every day.

Without adequate vitamins and minerals, the body cannot efficiently use the macronutrients consumed.

Vitamins

Vitamins are organic compounds that help regulate body processes.

Different vitamins serve different roles.

Examples include:

Vitamin A

Supports:

- Vision
- Skin health
- Immune function

Vitamin B Complex

Supports:

- Energy metabolism
- Brain health
- Nerve function
- Red blood cell production

Vitamin C

Supports:

- Immune function
- Collagen production
- Tissue repair
- Antioxidant protection

Vitamin D

Supports:

- Bone health
- Immune regulation
- Calcium absorption

Vitamin E

Supports:

- Cellular protection
- Healthy skin

- Antioxidant activity

Vitamin K

Supports:

- Blood clotting
- Bone metabolism

Minerals

Minerals are inorganic nutrients required for normal body function.

Some are needed in larger amounts, while others are needed only in trace quantities.

Major Minerals

Examples include:

- Calcium
- Magnesium
- Potassium
- Sodium
- Phosphorus

These minerals support:

- Muscle function
- Bone strength
- Hydration
- Nerve communication
- Heart rhythm

Trace Minerals

Examples include:

- Zinc
- Iron
- Selenium
- Copper

- Iodine

These minerals support:

- Immune health
- Thyroid function
- Oxygen transport
- Enzyme activity
- Antioxidant defenses

Nutrient Density

Holistic nutrition emphasizes choosing foods that provide the greatest amount of nutrition with the fewest highly processed ingredients.

Nutrient-dense foods are rich in:

- Vitamins
- Minerals
- Fiber
- Healthy fats
- Protein
- Phytonutrients

Examples include:

- Leafy greens
- Colorful vegetables
- Berries
- Beans
- Nuts
- Seeds
- Whole grains
- Herbs

The more colorful and less processed the food, the greater its nutrient density tends to be.

Nutrient Balance

Holistic nutrition teaches that balance is more important than extremes.

No single food provides every nutrient.

Instead, good nutrition comes from eating a wide variety of wholesome foods over time.

Balance includes:

- Adequate protein
- Healthy fats
- Complex carbohydrates
- Fruits
- Vegetables
- Water
- Appropriate vitamin and mineral intake

Variety helps reduce nutritional gaps and supports long-term wellness.

Nutrient Deficiencies

When the body does not receive enough nutrients over time, normal function may become impaired.

Possible consequences of inadequate nutrition may include:

- Fatigue
- Poor concentration
- Slow healing
- Weak bones
- Reduced immune function
- Muscle weakness
- Dry skin
- Hair changes
- Digestive disturbances

Because nutrients work together, a deficiency in one area may affect many body systems.

Whole Foods First

One of the core principles of holistic nutrition is that food should be the primary source of nutrients whenever possible.

Whole foods naturally contain combinations of:

- Vitamins
- Minerals
- Fiber
- Antioxidants

- Phytochemicals

These nutrients work together in ways that supplements alone cannot fully replicate.

Supplements may be appropriate in certain situations but are generally intended to complement, not replace, a balanced diet.

Holistic Perspective

Rather than focusing on a single nutrient, holistic nutrition considers the entire dietary pattern.

Good nutrition is built through daily habits, including:

- Eating a variety of colorful foods
- Staying hydrated
- Choosing minimally processed foods
- Eating balanced meals
- Practicing moderation
- Supporting digestion
- Living an active lifestyle

Small, consistent choices made over time have a greater impact than occasional dramatic changes.

Key Terms

Students should become familiar with the following vocabulary:

- Nutrient
- Macronutrient
- Micronutrient
- Carbohydrate
- Protein
- Fat
- Amino Acid
- Vitamin
- Mineral
- Dietary Fiber
- Nutrient Density
- Essential Nutrient
- Balanced Nutrition

Lesson Summary

In this lesson, you learned that nutrients are the foundation of human health and are divided into two major categories: **macronutrients** and **micronutrients**. You explored how carbohydrates provide energy, proteins build and repair tissues, and healthy fats support hormones, brain function, and vitamin absorption. You also examined the essential roles of vitamins and minerals in regulating the body's many biochemical processes.

Finally, you learned that holistic nutrition emphasizes obtaining nutrients from a wide variety of whole, nutrient-dense foods while maintaining balance rather than relying on restrictive eating patterns. This knowledge prepares you to better understand how nutrition supports every body system, which will be explored in the upcoming lessons.

Required Reading

The Holistic Guide to Human Nutrition

- Chapter 3 – Understanding Macronutrients and Micronutrients

Holistic Nutrition Class

Suggested Reading:

- Human Nutrition
- Macronutrients
- Micronutrients
- Protein
- Carbohydrates
- Nutritional Science
- Nutrient Classes